Student's 't' statistics -

as it follows a distribution called Student t-distribution with pdf

$$f(t) = \frac{1}{\sqrt{\theta}\, B(\frac{1}{2}, \theta/2)} \frac{1}{\left(1+\frac{t^2}{\theta}\right)^{(\theta+1)/2}}$$

$-\infty < t < \infty$.

where $\theta = (n-1)$ ~~df~~ is df.

if $\theta = 1$. then

$$f(t) = \frac{1}{B(\frac{1}{2}, \frac{1}{2})} \frac{1}{(1+t^2)}$$

$$= \frac{1}{\pi} \frac{1}{(1+t^2)}$$

which is standard Cauchy's distribution.

$f(t)$ is symmetrical about the line $t = 0$.

$\therefore$ odd order ~~mom~~ raw moment is zero. $\therefore$ mean $\mu_1' = 0$.

$$\mu_{2r}' = \int_{-\infty}^{\infty} t^2 f(t)\, dt$$

$$= \frac{n^r (2r-1)(2r-3)\cdots 3\cdot 1}{(n-2)(n-4)\cdots(n-2r)}$$

$$\therefore \sigma^2 = \text{Variance} = \mu_2 = \frac{n}{n-2}.$$

mgf of t-distribution does not exist.